

AI Distribution Engine

Product Requirements Document — MVP v1.0

"Find where your customers already hang out."

Product: AI Growth Intelligence Platform

Stage: MVP — Reddit-First

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1. Executive Summary

AI Distribution Engine is a Reddit-first growth intelligence platform that helps SaaS founders, indie hackers, creators, and agencies discover where their target customers actively discuss problems, ask for recommendations, and express buying intent — automatically, via AI-powered analysis.

The platform ingests Reddit discussions relevant to a user's business niche, processes them through a dual-model Gemini AI pipeline (Flash-Lite for extraction, Flash for synthesis), and surfaces actionable growth opportunities: communities to engage, content ideas rooted in real conversations, and competitor blind spots to exploit.

Core Promise	"Find where your customers already hang out."
Product Type	AI Growth Intelligence Platform
MVP Scope	Reddit-only, three-tier SaaS (Free / Starter / Pro)
Target Market	Indian SaaS founders, indie hackers, creators, and agencies
Revenue Model	Monthly subscriptions via Razorpay (Rs.499 and Rs.999 tiers)
Primary KPI	Weekly Returning Users
Stack	Next.js + Tailwind (Vercel) Flask + Gunicorn (Railway) Supabase PostgreSQL Gemini API Reddit API

2. Product Vision & Positioning

What We Are Building

A Growth Opportunity Discovery Engine. The platform monitors Reddit's real-time discussions on behalf of its users and converts raw community activity into prioritised, AI-annotated opportunities — each with a suggested action, intent classification, and engagement score.

What We Are NOT Building

- ✗ An AI chatbot or generic assistant
- ✗ A social media scheduling or management tool
- ✗ A generic analytics or reporting dashboard
- ✗ A CRM, browser extension, or mobile app (for MVP)
- ✗ A LinkedIn, X, or YouTube analyser (for MVP)

Why Reddit First

Reddit offers unfiltered, keyword-searchable, high-intent conversations across thousands of niche communities. Unlike LinkedIn (professional posturing) or X/Twitter (noise-heavy), Reddit discussions surface genuine pain points, comparison requests, and buying signals. The Reddit API is also freely available with generous rate limits, keeping MVP infrastructure costs extremely low.

3. Target Users & Personas

SaaS Founder

Description	Building a B2B or B2C SaaS product. Needs distribution channels but lacks time for manual Reddit research. Values weekly automated intelligence over ad-hoc browsing.
Best Plan	Rs.999/mo Pro plan
Key Features	Competitor monitoring, weekly intelligence reports, priority AI

Indie Hacker

Description	Solo builder launching micro-SaaS products. Budget-conscious but highly motivated. Willing to pay Rs.499/mo if the opportunity feed saves 5+ hours/week.
Best Plan	Rs.499/mo Starter plan
Key Features	Opportunity feed, Reddit community discovery, saved opportunities

Content Creator

Description	YouTuber, newsletter writer, or course creator targeting a niche. Wants content ideas rooted in what their audience is actually asking.
Best Plan	Rs.499/mo Starter plan
Key Features	Content idea generation, audience pain-point discovery

Growth Agency

Description	Manages growth for 3–10 clients simultaneously. Needs competitor intelligence and opportunity feeds per client business profile.
Best Plan	Rs.999/mo Pro plan
Key Features	Multi-business profiles, competitor tracking, advanced scoring

Small Business Owner

Description	Running a local or online business. May not know Reddit well but wants to understand where their customers hang out online.
Best Plan	Free or Starter
Key Features	Basic report generation, subreddit discovery

4. Core Value Proposition

The platform's value is not data - it is the transformation of raw community activity into **specific, immediately actionable growth moves**. A user does not need to know what subreddits exist, how to write Reddit search queries, or how to interpret engagement metrics. They describe their business once; the engine handles the rest.

Without the Platform	With the Platform
Manually browse 20+ subreddits per week	Automated daily/weekly Reddit scanning
Miss competitor mentions in niche communities	Real-time competitor mention alerts (Pro)
Guess what content resonates with audience	AI-extracted content ideas from real conversations
No visibility on where buyers are discussing problems	Buying intent classification per post
Hours of research per growth campaign	5-minute weekly digest with top opportunities

5. Pricing & Plan Structure

FREE

Rs.0 / mo

- 1 report/month
- 1 competitor
- Limited opportunities
- Weekly digest

STARTER

Rs.499 / mo

- 20 reports/month
- 5 competitors
- Weekly opportunity feed
- Saved opportunities
- Advanced AI recommendations

PRO

Rs.999 / mo

- 50 reports/month
- Advanced competitor tracking
- Recurring monitoring
- Priority AI analysis
- Daily opportunity refresh
- Competitor overlap detection

Detailed Plan Comparison

Feature	Free	Starter (Rs.499)	Pro (Rs.999)
Reports / month	1	20	50
Competitors tracked	1	5	Unlimited
Opportunity feed	Limited (3)	Full (20+)	Full + Advanced
Opportunity refresh rate	Weekly	Every few days	Daily
Saved opportunities	Limited	Unlimited	Unlimited
Competitor analysis	Basic	Multiple	Recurring monitoring
Weekly digest	Yes	Yes	Yes
Email digest	Yes	Yes	Yes
Priority AI processing	No	No	Yes
Competitor overlap detect	No	No	Yes
Keyword trend tracking	No	No	Yes
Background monitoring	No	Starter feed	Full Pro monitoring
Advanced opportunity score	No	No	Yes

Billing Notes

- All billing is monthly via Razorpay subscriptions.
- Report counts reset on the user's monthly billing cycle date.
- Cancelled subscriptions downgrade to Free at end of billing period.
- Failed payments enter a 3-day grace period before plan downgrade.

- No annual plan in MVP (future roadmap item).

6. Feature Specifications

6.1 Landing Page (/)

Goal: Convert visitor to signup in under 10 seconds.

Sections & Requirements

Section	Content / Requirement
Hero	Headline: "Find where your customers already hang out." Subheadline: "Discover Reddit communities, growth opportunities, competitor strategies, and content ideas — automatically." Primary CTA: [Analyse My Business] → /login Secondary CTA: [View Demo Report] → demo modal
Social Proof	3 example insight cards: Opportunity Found, Competitor Insight, Content Opportunity — using realistic fake data
How It Works	3-step explainer: Describe Business → AI Scans Reddit → Receive Growth Intelligence
Pricing Preview	Compact 3-tier pricing with CTAs
Footer	Privacy Policy Terms of Service Contact

6.2 Authentication (/login)

Requirement	Detail
Login methods	Google OAuth (via Supabase) + Email/Password
Post-login logic	New user → /onboarding Returning user → /dashboard
Error handling	Invalid credentials shown inline, not in alert()
Forgot password	Email reset via Supabase Auth
Session storage	Supabase JWT stored in cookie (HttpOnly where possible)

6.3 Onboarding (/onboarding)

Three-step guided setup to collect business context before first report.

Step	Fields / Actions
Step 1 — Business Info	Business Name (required) Website URL (optional) Category (dropdown: SaaS/Creator Tool/Agency/AI Tool/Ecommerce/Local Business/Other) Target Audience (free text) Primary Goal (radio: Leads/Traffic/Brand Awareness/Audience Growth/Sales) Region (select: India/Global/USA/UK/Other)
Step 2 — Competitors	Add up to plan limit (Free: 1, Starter: 5). Display as removable chips. CTA: [Analyse Competitors]
Step 3 — Platform Select	Reddit (enabled, pre-ticked). LinkedIn/X/YouTube/IndieHackers shown as Coming Soon. CTA: [Generate Growth Report] → triggers first report

6.4 Report Generation Engine

The core AI pipeline. Triggered by user action or background jobs.

Step	Action	Model / Service
1	Validate user has reports remaining	Backend (usage_service)
2	Fetch business + competitor context	Supabase
3	Expand business description into search keywords	Gemini Flash-Lite
4	Search Reddit for posts + top comments	Reddit API (PRAW)
5	Summarise raw Reddit posts in bulk	Gemini Flash-Lite
6	Extract & classify growth opportunities	Gemini Flash-Lite
7	Generate full strategy report (JSON)	Gemini Flash
8	Store report + opportunities in Supabase	Supabase
9	Increment monthly report count	usage_tracking

Loading UX

Animated 5-step progress indicator shown during generation (15–40 sec estimate): Finding communities → Analysing competitors → Detecting opportunities → Extracting pain points → Building growth strategy.

6.5 Main Dashboard (/dashboard)

Section	Content
Layout	Left sidebar nav + top header bar + main content area
Sidebar items	Dashboard Opportunities Competitors Reports Saved Billing Settings
Top header	Plan badge Reports remaining counter Upgrade CTA (hidden for Pro)
Growth Score card	Score /100 with 3 insight bullets (e.g. 'Strong Reddit demand')
Best Communities	Top 3 subreddit cards with: member count, activity level, suggested strategy
Opportunity preview	3 opportunity cards (plan-gated; Free sees blurred cards)
Competitor preview	1–3 competitor insight snippets
Content Ideas	3–5 AI-generated content titles based on Reddit discussions
Weekly Digest	Latest digest summary with CTA to full report

6.6 Opportunity Feed (/dashboard/opportunities)

Core retention feature. Users return weekly to discover new actionable opportunities.

Field	Description
Title	Reddit post title or AI-extracted insight title
Source	Subreddit name (r/videoediting, r/SaaS, etc.)

Engagement	Comment count + upvote score
Intent badge	Buying Signal / Pain Point / Comparison / Discussion
AI Insight	1-sentence AI summary of why this is relevant
Suggested Action	Specific recommended next step (e.g. 'Reply with case study')
Opportunity Score	0–100 AI-calculated relevance score
CTA: Save	Bookmark to Saved section (plan-gated)
CTA: Open	External link to Reddit thread

Plan Gating

Plan	Visible Opportunities	Refresh Rate	Features
Free	3 (blurred beyond 3)	Weekly	Basic fields only
Starter	Up to 20	Every few days	AI insights + actions + save
Pro	All	Daily	Advanced scoring + competitor overlap

6.7 Competitor Intelligence (/dashboard/competitors)

Plan	Features
Free	1 competitor, basic Reddit mention analysis
Starter	Up to 5 competitors, engagement pattern insights, underserved community gaps
Pro	Daily automated monitoring, content pattern detection, side-by-side comparison, activity spike alerts

Competitor Card Data

- Competitor name + website link
- Top performing content types on Reddit
- Reddit engagement score (posts mentioning competitor)
- Identified underserved communities (gaps the competitor ignores)
- Growth opportunities vs. this competitor
- (Pro) Last monitored date + activity trend arrow

6.8 Reports (/dashboard/reports)

Report detail view renders the full JSON report_data into readable sections: Growth Score, Best Communities, Extracted Opportunities, Content Ideas, and Strategic Recommendations. Users can delete old reports or generate new ones (subject to monthly limit).

6.9 Saved Opportunities (/dashboard/saved)

Bookmarked opportunities persist across sessions. Free plan has a limited save count (3–5 items). Starter and Pro have unlimited saves. Items can be filtered by intent type or subreddit.

6.10 Weekly Digest

Field	Content
Trigger	Every Monday 6:00 AM IST (background job)
Delivery	In-app dashboard widget + email
Plans	All plans receive digest
Content	New discussions found Rising keywords Competitor trends Best subreddit this week Recommended action
Sample headline	"AI creator tools discussions increased 42% this week"

6.11 Billing & Subscription (/pricing, /checkout, /billing/*)

Flow	Steps
Upgrade	1. User clicks upgrade CTA 2. Redirected to /checkout?plan=starter pro 3. Backend creates Razorpay order 4. Razorpay checkout modal opens 5. User pays 6. Backend verifies HMAC signature 7. subscriptions table updated 8. users.plan updated 9. Redirect to /billing/success
Cancellation	1. User clicks Cancel in /billing/manage 2. Confirm dialog 3. Backend cancels Razorpay subscription 4. Plan downgraded to Free at period end 5. Confirmation shown
Failed payment	1. Razorpay webhook fires payment.failed 2. 3-day grace period (plan stays active) 3. Retry notification sent 4. If still unpaid: downgrade to Free

7. System Architecture

High-Level Component Diagram

The architecture separates concerns cleanly: Next.js handles all UI and client-side logic, Flask serves as the stateless API layer orchestrating external services, and Supabase acts as both the authentication provider and the primary data store.

Layer	Technology	Hosting	Responsibility
Frontend	Next.js 14 App Router + Tailwind + shadcn/ui	Vercel	UI rendering, auth session management, API calls to Flask
Backend	Flask + Gunicorn	Railway	Business logic, Reddit + Gemini orchestration, billing, usage enforcement
Database	Supabase PostgreSQL	Supabase	Persistent data storage, Row Level Security, Auth
AI Models	Gemini Flash-Lite + Flash	Google	Keyword expansion, summarisation, opportunity extraction, strategy reports
Reddit API	PRAW (Python Reddit API Wrapper)	External	Post search, comment fetching, engagement data
Payments	Razorpay Subscriptions	External	Monthly billing, webhooks, subscription lifecycle
Jobs	APScheduler (in-process)	Railway	Weekly digest, opportunity scanner, competitor monitor

Request Flow: Report Generation

#	From	To	Action
1	Browser	Next.js	User clicks Generate Report
2	Next.js	Flask API	POST /reports/generate {business_id}
3	Flask	Supabase	Check usage_tracking (reports_used < limit)
4	Flask	Flask	Fetch business + competitor data
5	Flask	Gemini Lite	Generate keyword list from business description
6	Flask	Reddit API	Search posts by keywords across subreddits
7	Flask	Gemini Lite	Bulk summarise + extract opportunities
8	Flask	Gemini Flash	Generate final strategy report JSON
9	Flask	Supabase	Save report + opportunities + increment usage
10	Flask	Next.js	Return report data JSON
11	Next.js	Browser	Render report dashboard sections

8. Database Schema

Table: users

Column	Type / Constraint	Notes
id	UUID	Primary key
email	TEXT UNIQUE NOT NULL	User email
name	TEXT	Display name
plan	TEXT DEFAULT 'free'	free starter pro
avatar_url	TEXT	Profile photo URL
created_at	TIMESTAMPTZ	Account creation time

Table: businesses

Column	Type / Constraint	Notes
id	UUID	Primary key
user_id	UUID FK → users	Owner
business_name	TEXT NOT NULL	Company/product name
website	TEXT	Optional URL
category	TEXT	SaaS Creator Tool Agency etc.
target_audience	TEXT	Free-text description
goal	TEXT	Leads Traffic Brand Awareness etc.
region	TEXT	India Global USA etc.
created_at	TIMESTAMPTZ	

Table: competitors

Column	Type / Constraint	Notes
id	UUID	Primary key
business_id	UUID FK → businesses	
competitor_name	TEXT NOT NULL	
website	TEXT	
created_at	TIMESTAMPTZ	

Table: reports

Column	Type / Constraint	Notes
id	UUID	Primary key

business_id	UUID FK → businesses	
report_type	TEXT	growth competitor opportunity
report_data	JSONB NOT NULL	Full AI-generated report payload
status	TEXT	pending complete failed
created_at	TIMESTAMPTZ	

Table: opportunities

Column	Type / Constraint	Notes
id	UUID	Primary key
business_id	UUID FK → businesses	
title	TEXT	Post title or AI insight title
source	TEXT	reddit
subreddit	TEXT	e.g. r/SaaS
url	TEXT	Reddit thread URL
engagement_score	INTEGER	Upvotes + comments combined
opportunity_score	INTEGER	AI score 0–100
ai_summary	TEXT	1-sentence AI insight
recommended_action	TEXT	Specific suggested action
intent_type	TEXT	buying pain_point comparison discussion
created_at	TIMESTAMPTZ	

Table: saved_opportunities

Column	Type / Constraint	Notes
id	UUID	Primary key
user_id	UUID FK → users	
opportunity_id	UUID FK → opportunities	
saved_at	TIMESTAMPTZ	

Table: usage_tracking

Column	Type / Constraint	Notes
id	UUID	Primary key
user_id	UUID FK → users UNIQUE	One row per user
reports_used	INTEGER DEFAULT 0	Count this billing cycle

competitors_used	INTEGER DEFAULT 0	
monthly_reset_date	DATE	Next reset date

Table: subscriptions

Column	Type / Constraint	Notes
id	UUID	Primary key
user_id	UUID FK → users	
plan	TEXT	free starter pro
razorpay_customer_id	TEXT	
razorpay_subscription_id	TEXT	
status	TEXT	active cancelled past_due
renewal_date	DATE	Next billing date

Table: weekly_digests

Column	Type / Constraint	Notes
id	UUID	Primary key
business_id	UUID FK → businesses	
digest_data	JSONB	Weekly summary payload
created_at	TIMESTAMPTZ	

9. API Specification

All endpoints require **Authorization: Bearer {supabase_jwt}** except /health and /billing/webhook.

Auth

Method	Path	Description	Auth
POST	/auth/verify	Verify JWT + create user if new	Public
GET	/auth/me	Return current user profile + plan	Protected

Businesses

Method	Path	Description	Auth
POST	/businesses	Create a new business	Protected
GET	/businesses	List user's businesses	Protected
GET	/businesses/:id	Get single business	Protected
PUT	/businesses/:id	Update business	Protected
DELETE	/businesses/:id	Delete business + cascade	Protected

Competitors

Method	Path	Description	Auth
POST	/businesses/:id/competitors	Add competitor (limit enforced)	Protected
GET	/businesses/:id/competitors	List competitors	Protected
DELETE	/businesses/:id/competitors/:cid	Remove competitor	Protected

Reports

Method	Path	Description	Auth
POST	/reports/generate	Trigger report generation (limit enforced)	Protected
GET	/reports	List all reports (filter by business_id)	Protected

GET	/reports/:id	Get single report detail	Protected
DELETE	/reports/:id	Delete a report	Protected

Opportunities

Method	Path	Description	Auth
GET	/opportunities	List opportunities for a business (plan-gated)	Protected
POST	/opportunities/save	Save/bookmark an opportunity	Protected
DELETE	/opportunities/save/:id	Remove bookmark	Protected
GET	/opportunities/saved	Get user's saved opportunities	Protected

Usage

Method	Path	Description	Auth
GET	/usage	Get usage counts + limits for current user	Protected
GET	/usage/limits	Get plan limits (no usage data)	Protected

Billing

Method	Path	Description	Auth
POST	/billing/create-order	Create Razorpay order	Protected
POST	/billing/verify-payment	Verify payment + activate subscription	Protected
POST	/billing/webhook	Handle Razorpay webhook events	Public+HMAC
GET	/billing/subscription	Get current subscription details	Protected
POST	/billing/cancel	Cancel subscription	Protected

Misc

Method	Path	Description	Auth
GET	/health	Health check endpoint	Public

10. Background Jobs

Job	Schedule	Plans	Actions
Weekly Digest	Monday 6:00 AM IST	All	Scan Reddit trends → Gemini summary → Save digest → Send email
Opportunity Scanner	Every 4 hours	Starter + Pro	Scan Reddit → Score opportunities → Save new ones → Deduplicate
Competitor Monitor	Daily 8:00 AM IST	Pro only	Monitor competitor mentions → Detect engagement patterns → Alert on spikes

Job Failure Handling

Each job is wrapped in try/except. Failures are logged with timestamp and error message. A single job failure does not crash the APScheduler instance. Failed jobs are retried on the next scheduled run. Repeated failures (3+ in a row) trigger a Sentry alert.

11. AI Architecture & Prompts

Model	Tasks	Why
Gemini 1.5 Flash-Lite (Flash-Lite)	Keyword expansion Post summarisation Opportunity extraction Intent classification Keyword grouping	Extremely low cost per token. Ideal for bulk processing of large Reddit post volumes.
Gemini 1.5 Flash (Flash)	Final strategy report synthesis Competitor analysis Weekly digest generation	Higher reasoning quality for the user-facing output that drives retention.

Prompt Files

File	Purpose	Key Variables
opportunity_prompt.txt	Extract ranked opportunities from Reddit data	{business_name}, {category}, {target_audience}, {reddit_posts_summary}
competitor_prompt.txt	Analyse competitor Reddit presence + gaps	{competitor_name}, {website}, {reddit_mentions}
strategy_prompt.txt	Generate full growth strategy report (JSON output)	{business_context}, {opportunities_summary}, {competitor_context}

Caching Strategy (Cost Reduction)

Cache Target	TTL	Rationale
Subreddit search results	6 hours	Subreddit membership/activity changes slowly
Keyword lookup results	12 hours	Same keywords return same posts within a half-day
Gemini summaries (same input)	24 hours	Identical input = identical output; no re-inference needed
Competitor mention searches	6 hours	Competitor Reddit activity is not real-time sensitive

12. Subscription Enforcement Logic

Enforcement is applied at **two layers**: (1) frontend prevents the action from being initiated if the user is at their limit, and (2) backend independently verifies limits before executing any action. The backend is the source of truth.

Check Point	Frontend Behaviour	Backend Response
Report limit reached	Generate button disabled + tooltip 'Upgrade for more reports'	402 { error: 'limit_exceeded', upgrade_required: true, plan_needed: 'starter' }
Competitor limit reached	Add Competitor button disabled + tooltip	402 { error: 'competitor_limit', upgrade_required: true }
Pro-gated feature access	Feature shown with lock icon + upgrade CTA overlay	403 { error: 'plan_required', required_plan: 'pro' }
Monthly reset	Counter resets visually on billing date	Backend auto-resets reports_used on monthly_reset_date

13. Infrastructure & Hosting

Service	Provider	Plan	Estimated Monthly Cost
Frontend hosting	Vercel	Free Hobby / Pro	Rs.0 (Free tier sufficient for MVP)
Backend API hosting	Railway	Starter	Rs.500–Rs.2,000
Database + Auth	Supabase	Free tier	Rs.0 (Free tier: 500MB, 2GB bandwidth)
Gemini AI API	Google	Pay-per-use	Rs.500–Rs.3,000 (depends on usage)
Reddit API	Reddit	Free OAuth	Rs.0
Email (Resend)	Resend	Free tier	Rs.0 (100 emails/day free)
Error tracking	Sentry	Free tier	Rs.0
Total (0–100 users)	—	—	Rs.1,000–Rs.5,000 / month

14. Security Requirements

Requirement	Implementation
Authentication	Supabase Auth JWT. Verified on every backend request via service role key.
Row Level Security	Enabled on all Supabase tables. Users can only access their own data.
API Key Storage	All keys in environment variables. Never hardcoded. Never exposed to frontend.
CORS	Flask CORS restricted to Vercel production domain only.
Webhook Signature	Razorpay webhook HMAC-SHA256 signature verified before processing.
Rate Limiting	Flask-Limiter: 5 req/min on report generation, 60 req/min global per IP.
Input Sanitisation	All user text fields sanitised server-side before storage.
HTTPS	Enforced by Vercel (frontend) and Railway (backend) by default.
Secrets Rotation	All API keys rotated before public launch.
Reddit ToS Compliance	No commercial resale of Reddit data. Used only for insight generation.

15. Non-Functional Requirements

Category	Requirement	Target
Performance	Report generation time	< 45 seconds (P95)
Performance	Dashboard initial load	< 2 seconds (LCP)
Performance	API endpoint latency (non-AI)	< 300ms (P95)
Availability	Backend uptime	> 99% monthly
Availability	Frontend (Vercel)	> 99.9% (Vercel SLA)
Scalability	Concurrent report generations	10+ without queue (MVP)
Reliability	Background job success rate	> 95% per run
Usability	Mobile responsiveness	Fully responsive (iOS Safari, Android Chrome)
Usability	Onboarding completion	Target > 70% who start, complete Step 3
Cost	AI cost per report	< Rs.5 per full report at MVP scale

16. Success Metrics & KPIs

Primary KPI: Weekly Returning Users

The North Star metric is the percentage of registered users who return to the platform within any given 7-day window. A user who finds genuine value will return weekly to check new opportunities, view the digest, and monitor competitors. If weekly return rate is > 30% after the first month, the retention loop is working.

Metric	Target (Month 1)	Target (Month 3)	Measurement Tool
Weekly Return Rate	> 20%	> 35%	PostHog / Mixpanel
Onboarding Completion Rate	> 60%	> 70%	PostHog funnel
Report Generation Rate	> 1/user/week	> 2/user/week	Supabase usage_tracking
Free → Starter Conversion	> 3%	> 8%	Razorpay + Supabase
Starter → Pro Upgrade	—	> 15% of Starter	Razorpay
Monthly Churn Rate	< 15%	< 10%	Razorpay subscriptions
Opportunity Save Rate	> 10% of shown	> 20%	saved_opportunities table
MRR (Month 3 target)	Rs.10,000	Rs.50,000	Razorpay dashboard

17. MVP Boundaries — What NOT to Build

The following items are explicitly **out of scope** for the MVP. They represent future roadmap items. Building any of these before proving core retention would be premature.

Feature	Rationale for Exclusion
LinkedIn / X / YouTube channel integration	Additional API complexity and cost. Reddit is sufficient for MVP validation.
CRM integration	Not needed until users prove they're acting on opportunities at scale.
Social media scheduler / posting	Different product category. Adds complexity without proving core value.
AI agents / autonomous actions	High engineering cost. Out of scope until retention is proven.
Browser extension	Distribution complexity. Not needed for SaaS subscription model.
Mobile app (iOS / Android)	Web-first is sufficient for the target user (desk-based founders/agencies).
Team collaboration / multi-seat	Requires user management layer. Single-user first.
IndieHackers / Hacker News integration	Future channel after Reddit proves out.
Annual billing	Reduces initial revenue velocity. Monthly only for MVP.
AI-generated post drafts / scheduling	Separate product. Focus is intelligence, not content creation.

18. Development Phases

Phase	Focus	Key Deliverables	Completion Signal
Phase 1 Foundation	Core infrastructure	Auth (Supabase), Onboarding flow, Dashboard shell, Database schema, Supabase RLS	User can sign up, complete onboarding, and see empty dashboard
Phase 2 Data Layer	Reddit integration	Reddit API service (PRAW), Keyword search, Post + comment extraction, Engagement scoring	Backend can fetch 50+ relevant posts for any given business niche
Phase 3 AI Engine	Gemini integration	Dual-model pipeline, All 3 prompt files, Report generation endpoint, Opportunity extraction	Full report generated and stored in Supabase in < 45s
Phase 4 M onetisation	Billing + limits	Razorpay integration, Plan enforcement (frontend + backend), Upgrade modal, Usage tracking	User can pay Rs.499 and immediately see unlocked features
Phase 5 Retention	Recurring intelligence	Weekly digest job, Opportunity scanner job, Competitor monitor job, Email notifications	Digest lands in user inbox every Monday; opportunity feed refreshes automatically

19. Risk Register

Risk	Probability	Impact	Mitigation
Reddit API rate limits hit during report generation	Medium	High	Implement PRAW rate-limit handling + exponential backoff. Cache results aggressively. Warn user if Reddit is slow.
Gemini API latency spikes (>60s)	Low-Medium	High	Set 60s timeout. Return partial results on timeout. Show user a 'partial report' with retry option.
Gemini API costs exceed budget at scale	Medium	Medium	Flash-Lite for all bulk tasks. Aggressive caching (24h TTL). Monitor cost per report daily. Set billing alert at Rs.3,000/mo.
Razorpay webhook delivery failures	Low	High	Log all webhook events. Implement idempotency keys. Provide manual plan override in admin if needed.
Low conversion from Free to Paid	Medium	High	Test upgrade modal copy and timing. Add 'teaser' for locked features. Follow up with email sequence.
Reddit blocks PRAW access (TOS change)	Low	Critical	Monitor Reddit API announcements. Maintain Reddit API ToS compliance. Have Pushshift as fallback.
User data breach	Low	Critical	Supabase RLS enforced. Service role key backend-only. HTTPS everywhere. Regular security audit.
User churn due to low report quality	Medium	High	Iterate on Gemini prompts weekly. Gather user feedback in-app. A/B test prompt variations.